

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2ndYear 2nd Term Regular Examination, 2021
CE 2241
(Principles of Environmental Engineering)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable value for any missing data.

Section – A

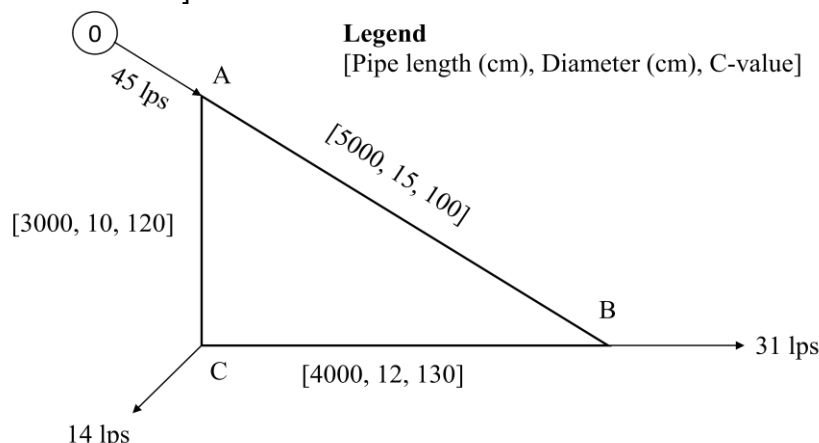
- 1 (a) What are the main objectives of water supply? Mention the basic elements of water supply with neat sketch. What are the responsibilities of water supply engineers? (14)
- (b) Define – i) Wholesome water ii) Palatable water ii) Potable water (09)
- (c) What are the factors affecting per capita water consumption? Mention the criteria required for water quality. (12)

- 2 (a) What is meant by 'Design period' and 'Population forecast'? Why population forecast is required in the design of public water supply scheme? (12)
- (b) What is water demand? Write down the factors that affect the rate of water demand. (08)
- (C) Estimate the population at 2030 and 2050 based on the given data. Use least square parabolic method. (15)

Year	1970	1980	1990	2000	210
Population (thousand)	200	215	240	270	325

- 3 (a) What is coagulation? Why coagulation and flocculation is accomplished simultaneously? Mention the key properties of coagulants. (10)
- (b) Define – i) Hindered settling ii) Break-point chlorination (06)
- (c) A sample of water contains following impurities. (06)
Source: River TC: 100 N/100 ml Color: 5 TCU
Arsenic: 0.007 mg/L Turbidity: 50 NTU
Draw the safest yet cheapest treatment process for the sample.
- (d) A rectangular settling tank without mechanical equipment is to treat 1.6 million liters per day of raw water. The sedimentation period is to be 4.1 hours. The flow velocity is 70mm/minute and depth of water and sediment is 4000mm. If an allowance of 1500 mm for sludge is made, determine the length and width of the settling tank. (13)

- 4 (a) What is disinfection? Draw the chlorine demand curve and briefly explain each zone with neat sketch. (08)
- (b) What is the difference between temporary and permanent hardness? Describe the base-exchange process of removing permanent hardness. (08)
- (c) Calculate the flow in each of the pipes of the following looped pipe network [at least two trials] (19)



Section – B

- 5 (a) What are the objectives of sanitation? Distinguish between on-site and off-site sanitation system. (08)
- (b) What is micro-organisms? Write down the role of micro-organisms in waste treatment. (08)
- (c) What is bacteria? Describe the bacterial growth kinetics with neat sketch. (11)
- (d) Explain the interrelationship between water, sanitation and health education. (08)
- 6 (a) What are differences between sewage and sullage? Write down the composition of sewage. (09)
- (b) Define BOD and COD. Why does dissolved oxygen (DO) in waste water decrease with the passing of time? (09)
- (c) Briefly explain the treatment steps of sewage treatment with neat sketch. (17)
- 7 (a) Distinguish between separate and combined sewerage system. What is self-cleaning and non-scouring velocity? (08)
- (b) Describe the technical advantages of Small Bore Sewers (SBS) system over the conventional sewerage with diagram. (10)
- (c) A septic tank is to be designed to serve a commercial building having 150 population and wastewater generation rate is 100 lpcd. The tank is to be desludged every five years. (17)
- 8 (a) What is solid waste management? Describe the functional elements of solid waste management showing their inter-relationship with a flow diagram. (11)
- (b) Discuss the following terms: i) Segregation, ii) Reduction, iii) Recycle, iv) Recycling, v) Resource recovery (15)
- (c) Describe the sources of air, water and soil pollution. (09)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 2nd Term Regular Examination, 2021
URP 2291
Statistics for Planners-II

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Necessary table/graphs/charts: (Graph Paper, t-table, χ^2 -table, Mann Whitney U table)

Section – A

1. (a) What is Hypothesis? It is generally hypothesized that “increasing income results in increasing private vehicle ownership”. Write the null and alternative hypothesis of the aforementioned working hypothesis. (10)
- (b) “Data in nature and collected randomly follow a normal distribution”-why? Explain with examples. (10)
- (c) What are the major checklists for a dataset to be normally distributed? A normality check for age distribution between treatment groups (0 and 1) found the following output for Kolmogorov-Smirnov and Shapiro-Wilk test. Is age normally distributed? - Explain. (15)

Test of Normality

	TRT	Kolmogorov-Smirnov			Shapiro-Wilk		
		Statistics	df	Sig.	Statistics	df	Sig.
age	0	0.108	14	0.200*	0.967	14	0.828
	1	0.124	11	0.200*	0.983	11	0.979

*This is a lower bound of the true significance

2. (a) How do you decide whether to conduct a parametric and non-parametric test based on variables? Make a list mentioning the advantages and disadvantages of non-parametric tests. (12)
- (b) The means of two random samples of sizes 9 and 7 are 94.4 and 96.8, and their standard deviations are 6.94 and 8.73 respectively. Can the sample be considered to have been drawn from the sample population? (15)
- (c) What is test of significance? Explain the concept of standard error. (08)
3. (a) Explain the properties of χ^2 (Chi-square) - test. (08)
- (b) Distinguish between sample and population. Mention the applications of Chi-square test in Urban Planning (10)
- (c) The following table gives the number of good and bad parts produced by each of three shifts in a factory: (17)

Shift	Good	Bad
Day	900	130
Evening	700	170
Night	400	200

Is there any association between the shift and the quality of parts produced? (use χ^2 - test)

4. (a) What is the basic difference between paired t-test and ANOVA tests? Explain the concepts of “between difference” and “within difference” with examples. (15)
- (b) A coaching center claims that a short course of one month can significantly improve the average marks in Mathematics course of class XII. A random sample of 12 students shows the following marks (out of 100) before and after one month coaching. (12)

Student	Before	After	Student	Before	After
1	78	85	7	52	68
2	65	66	8	63	69
3	85	89	9	68	75
4	74	68	10	36	47
5	36	48	11	42	52
6	45	45	12	40	48

Can we say that the claim of the coaching Center is true?

- (c) What is the difference between practical significance and statistical significance? How these two concepts apply to COVID-19 vaccine? (08)

Section – B

5. (a) “Correlation does not necessarily imply functional relationship”-Explain with examples. (10)
- (b) Discuss the scatter-diagram method of measuring correlation with necessary examples and sketches. (13)
- (b) Elucidate Lag and Lead of a correlation. Write down the assumptions of Karl Pearson’s Coefficient of correlation. (12)
6. (a) Using the following data: (25)
- Calculate the coefficient of correlation
 - Estimate the percentage of the group with lung cancer in a country, where 15% of the group smokes heavily.

Country	% of group smokes heavily	% of group with lung cancer
A	10	5
B	20	15
C	20	20
D	30	25
E	30	20
F	40	30

- (b) Discuss coefficient of determination in brief. State the utility of Regression in Urban Planning. (10)
7. (a) The following figures relate to length of service and income of the employees of an organization: (20)

Length of Service (Years)	11	7	2	5	8	6	10
Income (BDT. Thousands)	7	5	3	2	6	4	8

Compute the coefficient of correlation for the above data. Find the two regression equations and examine the relationship.

- (b) Explain seasonality in data. Why is it important to deseasonalize time series data?-explain with examples. (15)
- 8 (a) What is time series analysis? Illustrate the role of time series analysis in the decision making process of a planning project. (10)
- (b) Narrate the components of time series with necessary diagrams. (10)
- (c) The sales of a company in lakhs of BDT. for the year 1999 to 2005 are given below: (15)

Years	1999	2000	2001	2002	2003	2004	2005
Sales (in tonnes)	32	47	65	92	132	190	275

Estimate the sales figure for the year 2010 using an equation of the form $y=ab^x$.

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 2nd Term Regular Examination, 2021

URP-2281
(GIS and Remote Sensing)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) "A map is worth a thousand words." - Do you agree with the statement? State your rationale for your answer, using a spatial planning problem in your community. (15)
(b) Discuss the applicability of GIS to different aspects (i.e., location, condition, trend, routing, pattern, and modelling) to solve spatial planning problems in your city. (20)
2. (a) What is georeferencing? Explain the process of georeferencing for raster data using ArcGIS (10)
(b) Describe the concept of map overlay and its functions in GIS in solving the spatial problems. (15)
(c) Explain the different types of error produced while digitizing with necessary figures. (10)
3. (a) Explain three rules of topology with necessary sketch. (10)
(b) Explain the equation "Spatial Reference = Datum + Projection + Coordinate system" with necessary example. (15)
(c) Explain the "Database Management System" in GIS with necessary example. (10)
4. (a) Write down the functions of ArcMap, ArcCatalog, ArcScene, ArcGlobe and ArcTool box. (15)
(b) Write down the importance of layers in GIS with a focus to the practical application in Urban Planning. (10)
(c) Why GIS as a separate profession is important and so hot? Explain with a focus to Bangladesh. (10)

Section – B

5. (a) Describe the types of resolution in remote sensing by providing proper example. (20)
- (b) Classify the different types of satellites. Differentiate between Geostationary and Geosynchronous satellites. (10)
- (c) Write down the different types of applications of photogrammetry with a focus to urban and regional planning in context of Bangladesh. (05)
6. (a) Explain the components of a remote sensing system with proper figure. (10)
- (b) Describe scattering and the types of scattering. (07)
- (c) Provide a proper diagram for the reflectance of the main earth surface elements. (08)
- (d) Explain Instantaneous Field of View (IFOV) and Angular Field of View (AFOV) with proper figure. Calculate the IFOV of the following sensor
Detector Pixel Length = 350; Detector Field of View = 50 (10)
7. (a) Outline the various resolution of the following satellites: (15)
- Landsat
 - Sentinel 2
 - MODIS
- (b) List the advantages and constraints of using UAV and Satellite imagery. (10)
- (c) Explain how Band Combinations help in remote sensing analyses. (10)
8. (a) Explain the following (**any five**): (25)
- i. Methods for intensity interpolation
 - ii. Geometric and radiometric distortions
 - iii. K-means
 - iv. ISO data
 - v. Random Forest
 - vi. Maximum Likelihood
 - vii. Feature Extraction
- (b) “Band Algebra helps in creating indices such as NDVI” – Elaborate. (10)
How are they helpful in remote sensing analysis?

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 2nd Term Examination, 2021

URP 2251
(Landscape Planning and Design)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Figures in the right margin indicate full marks.
ii) Answer any **three** questions from each section in separate script.

Section – A

1. (a) What is landscape? (05)
(b) Landscape planning and design is very important for a sustainable built environment of a city. Evaluate this statement in terms of Urban and Regional Planning. (15)
(c) Describe the terms with example- (15)
i) Landscape features ii) Landscape elements iii) Landscape principles.
2. (a) Draw a sketch showing different types of lines in a plan of a house. (10)
(b) Narrate the basic color schemes used in landscaping with necessary drawing and example. (10)
(c) Describe the Hydrologic Cycle with relationship of different elements in necessary drawing. (15)
3. (a) Discuss Carbon Cycle with appropriate drawing. (11)
(b) Discuss the plant characteristics and design solutions with necessary sketch on any two of the followings: (24)
i) Front entry ii) Along sidewalks iii) Under window.
4. Write short notes (any five): (35)
a) Symmetrical balance in landscape design b) Different tree forms in landscape c) Repetition in landscaping d) Proportion in plant and hardscape e) Significance of studying human ecology f) Characteristics of plants and animals g) Landscaping with native wildflowers and grasses.

Section – B

5. (a) What is landscape site? (05)
(b) Explain the site inventory and analysis process. (10)
(c) Write short notes on: (20)
i) Cloisters and Physic Garden ii) Millennial Landscapes iii) Site and Context
iv) View and Vista.
6. (a) Why is water considered as a significant resource in landscape planning? (05)
(b) How can we get the maximum benefit from water features in a landscape site? (10)
(c) Discuss about water edge landscape principles with necessary sketches and examples. (20)
7. (a) What is residential landscaping? (05)
(b) Explain the plants as an important feature in landscape planning and design. (10)
(c) Discuss different steps to develop a residential landscape with sketches. (20)
8. (a) What is Axis? (05)
(b) What are the properties of Axis? (10)
(c) Explain the various “Structure Pattern” in city scale landscape planning and design with sketches? (20)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 2nd Term Regular Examination, 2021
Hum 2217
(Public and Local Government Finance)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

- 1 (a) Discuss the growing importance of public finance in a view of the market failure. (10)
(b) Do you think, the operations of public finance are able to achieve maximum social welfare for the people? (10)
(c) Describe the increasing role of public finance in a developing country like Bangladesh. (15)
- 2 (a) Discuss the rationale of Government Expenditure? (15)
(b) Write down the sectors of Government Expenditure. (10)
(c) What are the sources of public revenue. (10)
- 3 (a) What is externality? Explain the effect of a negative externality in production. (20)
(b) Why does the free rider problem induce the government to provide public goods? (15)
- 4 (a) Write short notes on the followings (any four) (35)
 - i. Free-rider Problem.
 - ii. Public debt.
 - iii. VAT & Tariff.
 - iv. Private, Public and free goods.
 - v. Tax & fee.

Section – B

- 5 (a) Write down your understanding about deficit financing. (05)
(b) Discuss the advantages of deficit financing. (15)
(c) What are the economic implications of the growing volume of public debt in Bangladesh. (15)
- 6 (a) Distinguish between revenue budget and development budget. (10)
(b) Mention the characteristics of an Ideal Budget. (10)
(c) Discuss the steps of formulating national budget. (15)
- 7 (a) What are the tiers of local government in Bangladesh? (08)
(b) Discuss the roles of local government. (12)
(c) State the functional domains of local government of Bangladesh. (15)
- 8 (a) State the concept of Pouroshova/ Municipalities as a local Government. (08)
(b) Describe the power and functions of Pouroshova as a local government. (12)
(c) What is Union Parishad? Describe the structure and election system of Union Parishad as a local government. (15)